Latent Boundary Detection for Rollout Re-Anchoring in Frozen JEPA World Models

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***Abstract*—**

***Index Terms*—artificial intelligence, world models, representation learning**

I. Introduction

An agent that acts in the world benefits from predicting the consequences of its actions, and world models are learned for this purpose [1]. Joint-Embedding Predictive Architectures (JEPA) are a family of world models that predict latent embeddings instead of raw pixels, which avoids pixel reconstruction and keeps the model lightweight. Such models have been applied to image and video understanding (I-JEPA [2], V-JEPA [3]) and to robot control, where LeWM pairs the predictive model with a sampling-based planner that rolls out candidate action sequences in imagination and executes the best one, as in model predictive control [4]. Yet rollouts of a frozen world model drift from the real dynamics, and the error grows with every imagined step, so planning success falls with task distance on PushT [6].

To solve this problem, hierarchical planners such as H-JEPA plan at multiple timescales, but an extra model must be trained end-to-end on top of the world model [5]. Subgoal planners shorten the effective horizon, but the subgoals they generate can be spurious and can lower success [6]. A question remains: can boundary information replace subgoals in keeping the rollout error small? If it can, knowing when to stop a rollout and when to re-anchor it is enough.

To answer this question, a latent boundary detector is built. Three generic signals of the latent stream are measured. Increment energy reflects the magnitude of the latent change. Direction change reflects turns of the latent trajectory. Prediction novelty reflects the mismatch between the predicted next state and the observed one. These signals are segmented offline with an unsupervised changepoint method [7], and the segmentation is distilled into a lightweight causal head. During planning, the head reads the imagined rollout stream and the real observation stream. A boundary event stops the current plan and forces an immediate re-anchoring. The contributions are:

- A paired-task study verifying that the timing of stopping and re-anchoring controls planning success in frozen world models.
- A latent boundary detector ($\sim$3k parameters) that decides when to stop a rollout and re-anchor it, without task rules and without retraining the world model.
- An evaluation on PushT and OGB Cube against flat and fixed-interval baselines.

II. Related Work

III. Method

IV. Experiments

V. Conclusion

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